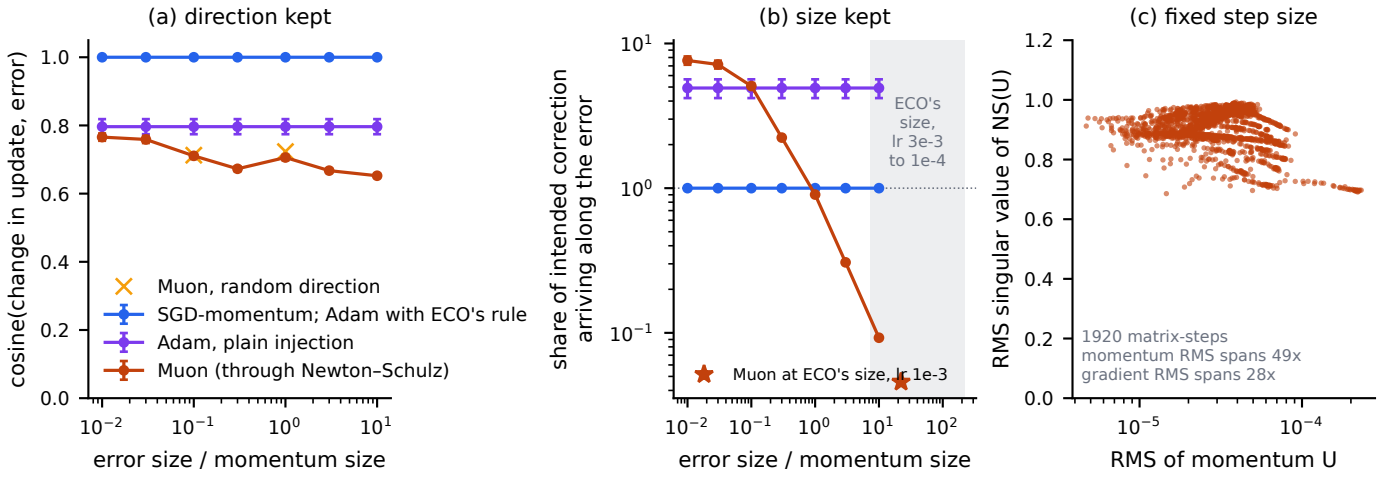

(a) direction kept
cosine(change in update, error)
1.0
0.8
0.6
0.4
0.2
0.0
Muon, random direction
SGD-momentum; Adam with ECO's rule
Adam, plain injection
Muon (through Newton–Schulz)
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
error size / momentum size
(b) size kept
share of intended correction arriving along the error
$10^{1}$
$10^{0}$
$10^{-1}$
ECO's size, lr 3e-3 to 1e-4
Muon at ECO's size, lr 1e-3
$10^{-2}$
$10^{-1}$
$10^{0}$
$10^{1}$
$10^{2}$
error size / momentum size
(c) fixed step size
RMS singular value of NS(U)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1920 matrix-steps
momentum RMS spans 49x
gradient RMS spans 28x
$10^{-5}$
$10^{-4}$
RMS of momentum U